

Objective:

To study MOSFET switching characteristics.

Theory:

A Metal Oxide Semiconductor Field Effect Transistor (MOSFET) is a voltage-controlled semi-conductor device that is widely used in electronic circuits. It has three main terminals: Gate (G), Drain (D) and Source (S). In the experiment, an IRF540 enhancement-type N-channel MOSFET is used as a switch.

The operation of the MOSFET depends on the gate-source voltage (V_{GS}), which is the potential difference between the gate and source terminals. When V_{GS} is lower than the threshold voltage (V_{th}), no conducting channel is formed between the drain and source. Therefore, the drain current (I_D) is almost zero, and MOSFET behave like an open circuit.

When V_{GS} becomes greater than the threshold voltage, a conducting channel is created between the drain and source. Current starts to flow through the MOSFET, and it behaves like a closed switch with very low resistance between the drain and source.

The switching operation of an enhancement NMOS MOSFET can be summarized as:

- If $V_{GS} < V_{TH}$:

- MOSFET is closed off.
- $I_D = 0$
- Drain and Source acts as open.

- If $V_{GS} > V_{TH}$:

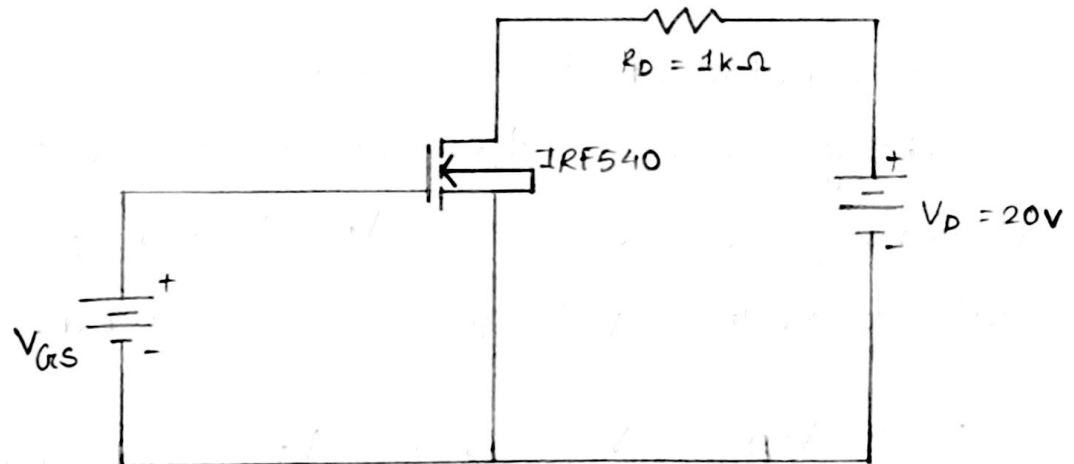
- MOSFET is on
- Current flows through drain and source.
- Drain and Source acts as a close switch.

Apparatus List:

Sl. No.	Component Details	Specification	Quantity
1	MOSFET	IRF540	1 piece each
2	Resistor	$1k\Omega$	1 piece each
3	POT		1 unit
4	Trainer Board		1 unit
5	DC Power Supply		2 unit
6	Digital Multimeter		1 unit
7	Chords and Wire		as required.

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Circuit Diagram:



Data Table:

$V_{DD} = 15V$				$V_{DD} = 20V$			
V_{GS}	V_{DS}	V_L	I_D	V_{GS}	V_{DS}	V_L	I_D
0V	14.99V	0V	0A	0V	19.94V	0V	0
0.5V	14.99V	0V	0A	0.5V	19.94V	0V	0
1V	14.99V	0V	0A	1V	19.94V	0V	0
1.5V	14.99V	0V	0A	1.5V	19.93V	0V	0
2V	14.99V	2.9mV	2.9μA	2V	19.93V	2.5mV	2.5μA
2.5V	14.97V	21.5mV	21.5μA	2.5V	19.91V	26.8mV	26.8μA
3V	14.56V	0.423V	0.42mA	3V	19.21 175mV	0.727V	0.72mA
3.5V	43.2mV	14.9V	14.9mA	3.5V	175mV	19.75V	19.75mA
4V	2.5mV	14.98V	14.98mA	4V	3.7mV	19.92V	19.92mA
4.5V	1mV	14.98V	14.98mA	4.5V	1.2mV	19.93V	19.93mA
5V	0.7mV	14.98V	14.98mA	5V	0.9mV	19.93V	19.93mA
6V	0.5mV	14.98V	14.98mA	6V	0.6mV	19.93V	19.93mA
7V	0.4mV	14.98V	14.98mA	7V	0.6mV	19.93V	19.93mA
8V	0.3mV	14.98V	14.98mA	8V	0.5mV	19.94V	19.94mA

Result Analysis

The objective of this experiment was to study the switching characteristics of an IRF540 MOSFET by observing the changes in V_{DS} , V_L and I_D as the gate-to-source voltage was increased.

From $V_{DD} = 15V$, when V_{GS} was varied from $0V$ to $1.5V$, the drain current (I_D) remained $0A$, the V_L was $0V$, and the V_{DS} stayed close to $14.99V$. This indicates that the MOSFET was in the off state, behaving like an open circuit switch.

At $V_{GS} = 2V$, a very small current of $2.5 \mu A$ was observed, showing that the MOSFET had started to turn on. As V_{GS} increased to $3V$, the drain current increased to $0.42 mA$, while V_{DS} decreased and V_L increased. At $V_{GS} = 3.5V$, the I_D rose to $14.9 mA$ and V_{DS} dropped to only a few mV. From $4V$ to $8V$, the current remained constant at $14.98 mA$, indicating that the MOSFET was fully on and the current was limited mainly by the $1 k\Omega$ load resistor.

For $V_{DD} = 20\text{ V}$, a similar switching behavior was observed. From 0 to 1.5 V, the drain current remained 0 A and V_{DS} was approximately 19.94 V, confirming the MOSFET was off.

When V_{GS} reached 3.5 V, the drain current increased rapidly to about 19.75 mA, while V_{DS} decreased to around 175 mV. For $V_{GS} > 4\text{ V}$, the current remained almost constant at 19.93 mA and V_{DS} was reduced to only a few mV, showing that the MOSFET was fully conducting.

The experiment results clearly demonstrate that the MOSFET operates as an electronic switch. Therefore, the observed results are consistent with the expected switching characteristics of an enhancement type NMOS MOSFET.

Question Answer:

Report: Describe the circuit operation using experimental data.

The circuit operated as an NMOS switch controlled by the gate-source voltage (V_{GS}). For both $V_{DD} = 15\text{ V}$ & $V_{DD} = 20\text{ V}$, when V_{GS} was between 0 V to 1.5 V , the I_D remained 0 A , while V_{DS} was almost equal to the supply voltage, indicating that the MOSFET was off. As V_{GS} increased to about $2\text{--}3\text{ V}$, the drain current started to increase, showing that the MOSFET had begun conducting. At $V_{GS} = 3.5\text{ V}$, the current increased rapidly to about 14.9 mA (15 V supply) and 19.75 mA (20 V supply), while V_{DS} dropped to a low value, indicating MOSFET is on. Therefore the experiment data confirms that MOSFET is an effective electronic switch that remains off below the threshold voltage and turning on when V_{GS} exceeds the threshold voltage.

Discussion:

This experiment helped us understand the switching characteristics of an ~~enh~~ enhancement type NMOS MOSFET. It was observed that when V_{GS} was below the threshold ~~frequency~~ voltage, the MOSFET remained off with almost ~~zero~~ zero drain current.

As V_{GS} increased beyond the threshold, the drain current increased while V_{DS} decreased, indicating that the MOSFET turned on and behaved like a closed switch. The experimental results agreed with the expected theoretical behavior.

Small differences between the theoretical and practical results may have occurred due to resistor tolerance, variations in the MOSFET Th. voltage, instrument accuracy and minor wiring or human errors during the experiment. Overall, the experiment successfully demonstrated the switching operation of an IRF540 MOSFET and improved our understanding of its practical application as an electronic switch.

Lab 6: Study of Switching Characteristics

Tahmid Mostofa
ID: 2524584043

